

Dr. Kathryn I. Wheeler

Williamstown, MA • kwheelerecology@gmail.com • <https://k-wheeler.github.io> • <https://github.com/k-wheeler>

EDUCATION

Boston University

Boston, MA

Doctor of Philosophy in Earth and Environment (GPA: 3.95)

Jan. 2023

Certificate in Biogeosciences

Dissertation: "Cold-deciduous broadleaf phenology: monitoring using a geostationary satellite and predicting using trigger-less dynamic models"

Advised By: Dr. Michael Dietze

Academic Awards: National Science Foundation Graduate Research Fellowship and Dean's Fellowship

Relevant Courses: Remote sensing, multivariate statistics, Bayesian statistics, ecological forecasting, terrestrial carbon

University of Delaware

Newark, DE

Bachelor of Science in Environmental Science (GPA: 3.95)

May 2017

Honors with Distinction, summa cum laude

Academic Awards: Full four-year merit scholarship, NOAA Hollings Scholarship, UD American Association of University

Professors Outstanding Senior, and Outstanding Senior in Environmental Science

Relevant Courses: GIS, soil science, plant physiology, ecosystem ecology, computer science, data structures

RELEVANT EXPERIENCE

Industrial Economics, Incorporated

Remote and Cambridge, MA

Environmental Consulting Associate

June 2023 – Present

- Produce technical analyses via Python, R, and ArcGIS Pro that quantify environmental damages at Superfund sites under Natural Resource Damage Assessment, directly informing legal decision-making
- Apply spatial and spatial-temporal statistical models (IDW, distributed-lag time series, mixed effects, generalized estimating equations, NMF, and PCA) to estimate contaminant impacts on fish populations under contested, high-stakes review
- Communicate quantitative findings to non-technical decision-makers (agency staff, clients, and attorneys) through written reports, presentations, and interactive tools
- Identify patterns in complex datasets using machine learning (decision trees and k-means clustering)
- Saved 120+ hours by using Python to automate running a client-preferred point-and-click application

Massachusetts Institute of Technology

Cambridge, MA

NOAA Climate and Global Change Postdoctoral Fellow

Sep. 2022 – June 2023

- Funded through a highly competitive 2-year fellowship (\$200k) selected through a written research proposal
- Led a multidisciplinary team of 43 to conduct ecological forecasts of tree leaf growth timing
- Assessed climate-change impacts on tropical forest carbon under drought using MODIS and Google Earth Engine
- Used LiDAR and eddy covariance flux data to investigate relationship between forest structure and productivity
- Wrote Bayesian and random forest machine learning models to analyze impacts of soil and biota on leaf seasonality

Boston University

Boston, MA

Research Assistant and Teaching Fellow

Sep. 2017 – Aug. 2022

- Created novel Bayesian statistical and mechanistic models in R to predict tree leaf seasonality
- Conducted field work at Harvard Forest to investigate the link between senescence and photosynthesis
- Developed automation pipeline to download, clean, organize, and condense 30 TB of satellite data from MODIS and GOES using a high-performance distributed system
- Led weekly lab based in R for 30 students in an environmental modeling course (Spring 2022)

University of Delaware

Newark, DE

Research Assistant and Computer Science Teacher Assistant

Aug. 2014 – May 2017

- Analyzed water quality data using MATLAB to fit advanced statistical and neural network models to available data
- Presented research at AGU and won two student presentation awards in hydrology and biogeoscience (top 5%)
- Provided teaching assistance for a weekly lab based in Python and MATLAB

INDEPENDENT PROJECTS

Oregon Land Cover Classification, Change, and Carbon Loss ([GitHub repo](#)) 2026

- Built an end-to-end geospatial machine learning pipeline classifying land cover and change across Oregon from satellite embeddings from Google Earth Engine
- Compared and tuned tree-ensemble, multilayer perceptron, and convolutional neural networks
- Designed a two-head neural network that predicts land-cover change directly from paired before/after embeddings
- Containerized the trained model with Docker and deployed it for wall-to-wall inference on Google Cloud Platform
- Ran CCDC and COLD algorithms to identify timing of land cover change
- Applied design-based statistical estimation (Olofsson accuracy assessment) to convert raw map output into bias-corrected area and carbon-loss estimates with confidence intervals

Massachusetts Appalachian Trail Phenology ([GitHub repo](#)) 2026

- Built an end-to-end machine learning pipeline predicting daily fall foliage phenological state for ~15,000 forest pixels, ingesting NASA HLS satellite imagery via Google Earth Engine
- Trained, tuned, and compared decision-tree classifiers and recurrent neural network (LSTM, PyTorch) to classify per-pixel state
- Deployed project as a self-updating webpage on my website: a scheduled pipeline (GitHub Actions) re-fetches imagery, recomputes predictions, and publishes an interactive map automatically every day
- Created the full stack, turning a research idea into a live, continuously updating product

SELECT PUBLICATIONS*

1. **Wheeler** et al. (2024). "Predicting spring phenology in deciduous broadleaf forests: NEON Phenology Forecasting Community Challenge." *Agr Forest Meteorol* 345: 109810.
2. **Wheeler** & Dietze (2021). "Improving the monitoring of deciduous broadleaf phenology using the Geostationary Operational Environmental Satellite (GOES) 16 and 17." *Biogeosciences* 18: 1971-1985.
3. Iida* & **Wheeler*** et al. (2021). "Canopy structure metrics governing stemflow funneling differs between leafed and leafless states: Insights from a large-scale rainfall simulator". *Hydrol Processes* 35:e14294.
*These authors contributed equally to this paper
4. **Wheeler** et al. (2020). "Visible and near-infrared hyperspectral indices explain more variation in lower-crown leaf nitrogen concentrations in autumn than in summer." *Oecologia*. 192:13-27.
Award: Highlighted Student Paper
5. **Wheeler** & Dietze (2019). "A statistical model for estimating midday NDVI from the Geostationary Operational Environmental Satellite (GOES) 16 and 17." *Remote Sensing* 11(21):2507.
6. **Wheeler** et al. (2017). "Tracking senescence-induced patterns in leaf litter leachate using parallel factor analysis modeling (PARAFAC) and self-organizing maps." *J. Geophys Res Biogeosci* 122(9):2233-2250.
7. Preprint: **Wheeler** & Dietze (2023). "A trigger may not be necessary to cause senescence in deciduous broadleaf forests." *bioRxiv* 10.1101/2023.06.07.544057.

*Full list on [Google Scholar](#)

SKILLS AND INTERESTS

Programming Languages: R, Python, MATLAB

Geospatial Tools: spatial R packages (sf, terra, raster), spatial Python packages (geopandas), Google Earth Engine, ArcGIS, spatial and spatial-temporal Inverse Distance Weighting

High-Performance Computing Tools: Git, shared clusters, parallelization, shell scripting (Unix, Bash), Slurm and Open Grid schedulers, Google Cloud Platform, Docker

Data Analysis Knowledge: Bayesian statistics (MCMC, state space, hierarchical models), frequentist statistics (distributed lag time series, Principal Component Analysis, mixed effects models, generalized estimating equations, non-negative matrix factorization, generalized additive models), machine learning (decision trees, tree ensembles, self-organizing maps, clustering, RNNs, CNNs, MLP), power analysis, sensitivity analysis, propagating and analyzing uncertainty, model assessment, data fusion and assimilation (Kalman Filter, Extended Kalman Filter, Ensemble Filters, Particle Filters)

Domain Knowledge: remote sensing, ecological forecasting, geospatial analysis, environmental modeling, forest ecology, carbon cycle, data science, AI capabilities (Claude Code)

Interests: running, hiking, cycling, backpacking, and reading